GOVERNMENT OF WESTERN AUSTRALIA

GSWA Geochronology Record

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Department of Mines, Industry Regulation and Safety

242455.1.pdf *Geochronology Record 1919*

242455.1: metagranodiorite, Mardoonganna Hills

(Mindoolah Tonalite, Thundelarra Supersuite; Youanmi Terrane, Yilgarn Craton)

SS Romano, Y Lu, IOH Fielding and TJ Ivanic

Location and sampling

BELELE SG50-11, KALLI 2344 Zone 50, MGA 541661E 7018077N
Warox Site TJIKAL190040 Sampled on 1 September 2019

This sample was collected from an outcrop about 7.8 km west-southwest of Mindodah Well, 5.9 km east-southeast of Sahara Well, and 5.2 km south of Mardoonganna Hills.

Geological context

The unit sampled is a metagranodiorite (Fig. 1) of the Mindoolah Tonalite of the Thundelarra Supersuite (Ivanic et al., 2022). The Mindoolah Tonalite intruded the 2979–2969 Ma felsic volcanics-dominated Madoonga Formation in the Weld Range area (Ivanic et al., 2022). A metarhyolite of the Madoonga Formation, collected about 5.8 km to the southeast, yielded an igneous crystallization age of 2977 ± 3 Ma (GSWA 155569, Wingate et al. 2019). A metarhyolite clast derived from the Madoonga Formation, collected about 10.6 km to the east-southeast, yielded an igneous crystallization age of 2979 ± 3 Ma (GSWA 193972, Wingate et al. 2012). A metasandstone of the Madoonga Formation, collected about 10.1 km to the east-southeast, yielded a single age population with maximum depositional age of 2969 ± 3 Ma (GSWA 184112, Wingate et al. 2008). Together with the geochronological data from the Madoonga Formation, the results from the current sample imply that a large area of c. 2.97 Ga crust is exposed in the Weld Range area.

Figure 1. Hand specimen image for sample 242455: metagranodiorite, Mardoonganna Hills

Petrographic description

The sample is a fine-grained, weakly porphyritic, and spherulitic granophyric metagranodiorite, consisting of about 40–45% plagioclase, 40% quartz, 10–15% K-feldspar, 4–5% biotite/chlorite, and accessory muscovite, apatite and zircon. Plagioclase occurs as sericitized, subhedral, and tabular grains that are mostly <1 mm in size with some less sericitized grains forming the finer quartz–feldspar groundmass. Quartz occurs as anhedral to subhedral grains up to 2 mm in size, a few with lobate grain boundaries. Smaller quartz grains are abundant in the quartz–feldspar groundmass, and in the granophyric patches. K-feldspar mainly occurs in the round, granophyric lenses, which range up to 2.5 mm in diameter. Brown biotite is anhedral to subhedral, forming laths up to 0.5 mm in size. Smaller biotite grains (<0.1 mm) occur

throughout the quartz–feldspar groundmass. Anhedral muscovite up to 0.2 mm is rare. Apatite mainly occurs as euhedral grains to 0.05 mm in length with a few grains up to 0.2 mm long. The texture of this rock is consistent with it being a high-level intrusive rock. It lacks a tectonic fabric but appears to have undergone some quartz recrystallization at greenschist facies conditions.

Zircon morphology

Zircons isolated from this sample are colourless to light brown and mainly subhedral to euhedral. The crystals are up to 300 µm long, with aspect ratios up to 4:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and many grains contain inclusions. A CL image of representative zircons is shown in Figure 2.

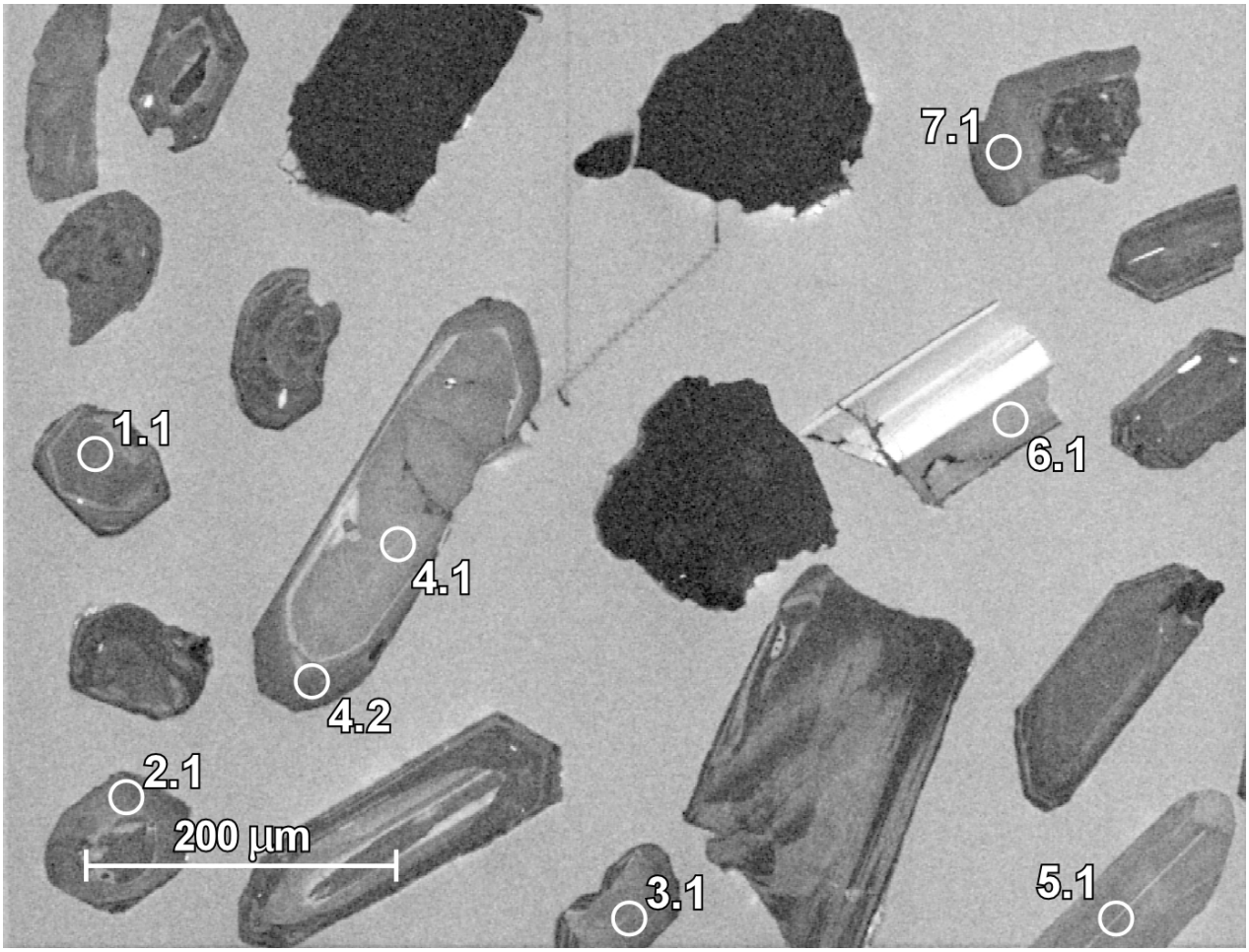

Figure 2. Cathodoluminescence image of representative zircons from sample 242455: metagranodiorite, Mardoonganna Hills. Numbered circles indicate the approximate locations of analysis sites

Analytical details

This sample was analysed on 23–25 January 2023, using SHRIMP-B. Seventeen analyses of the M257 standard were obtained during the session, of which 14 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.59% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.24% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1 (provided in the Links section). Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Fifteen analyses were obtained from 14 zircons. Results are listed in Table 1 and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3), and form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 15 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2973 ± 4 Ma (MSWD = 1.2).

The date of 2973 ± 4 Ma for the 15 analyses in Group I is interpreted as the igneous crystallization age of the protolith to the metagranodiorite.

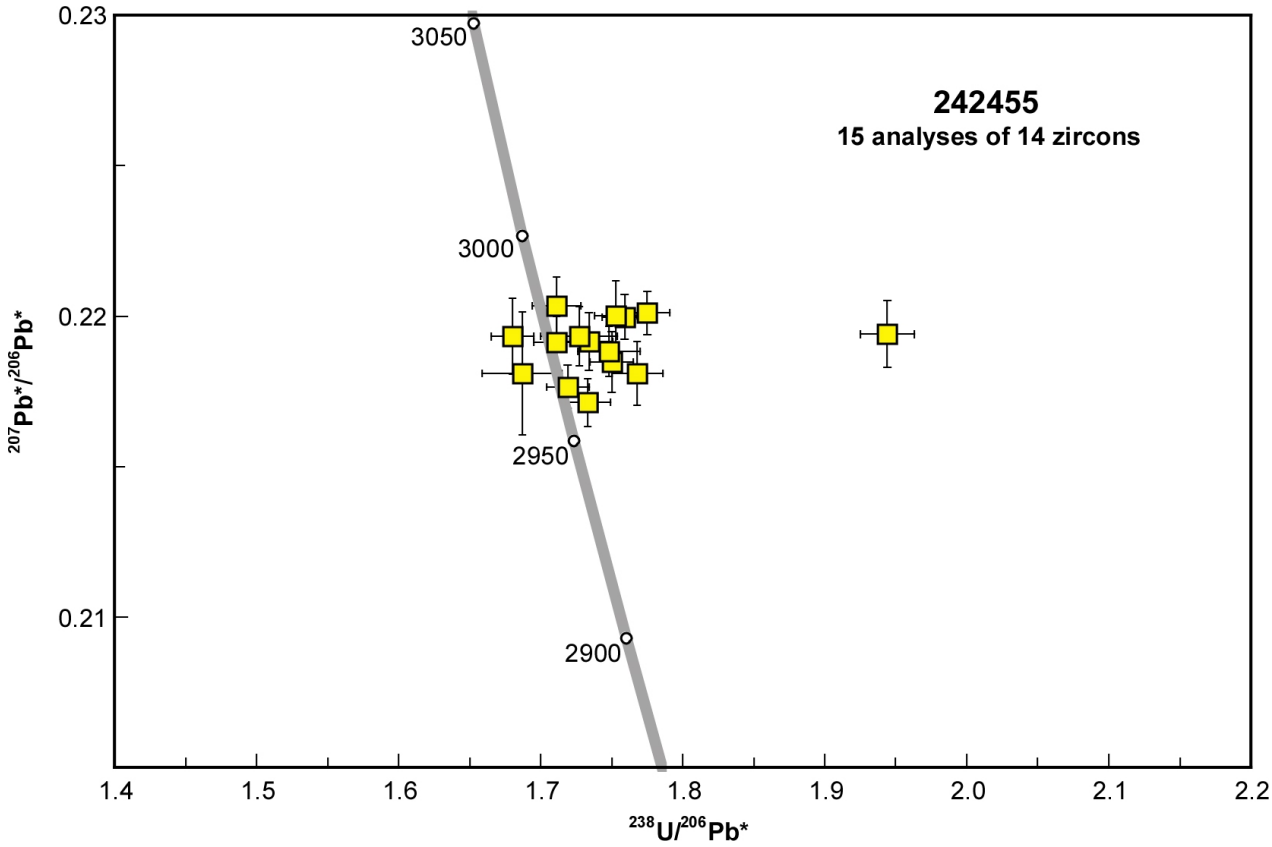

Figure 3. U–Pb analytical data for zircons from sample 242455: metagranodiorite, Mardoonganna Hills. Yellow squares indicate Group I (magmatic zircons)

Links

Table 1. Ion microprobe analytical results for zircons from sample 242455: metagranodiorite, Mardoonganna Hills: 242455.1.Table1.xls
SHRIMP spot images: 242455_spots_new.pdf
Introduction to geochronology information: Intro_2023.pdf

References

Ivanic, TJ, Wingate, MTD, Lowrey, JR and Lu, Y 2022, Formation of the Yilgarn protocraton by rift-related magmatism from 3.01 to 2.92 Ga: Geological Survey of Western Australia, Report 232, 34p.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Wingate, MTD, Bodorkos, S and Kirkland, CL 2008, 184112: metasandstone, Ram Well; Geochronology Record 736: Geological Survey of Western Australia, 6p. View Reference online

Wingate, MTD, Kirkland, CL and Ivanic, TJ 2012, 193972: metarhyolite clast in volcaniclastic breccia, Weld Range; Geochronology Record 1011: Geological Survey of Western Australia, 4p., <www.dmirs.wa.gov.au/geochron>. View Reference online

Wingate, MTD, Kirkland, CL, Van Kranendonk, MJ and Wyche, S 2019, 155569: metarhyolite, Weld Range; Geochronology Record 1096: Geological Survey of Western Australia, 4p., <www.dmirs.wa.gov.au/geochron>. View Reference online

Recommended reference for this publication

Romano, SS, Lu, Y, Fielding, IOH and Ivanic, TJ 2023, 242455.1: metagranodiorite, Mardoonganna Hills; Geochronology Record 1919: Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract viewed 11 October 2023. <www.dmirs.wa.gov.au/geochron>

Data obtained: 25 January 2023
Date released: 31 October 2023

This page was last modified on 11 October 2023.

Grid references in this publication refer to the Geocentric Datum of Australia 2020 (GDA20). All locations are quoted to at least the nearest 100 m.

Capitalized names in text refer to standard 1:100 000 map sheets, unless otherwise indicated.

WAROX is GSWA's field observation and sample database. WAROX site IDs have the format 'ABCXXXnnnnnnSS', where ABC = geologist username, XXX = project or map code, nnnnnn = 6-digit site number, and SS = optional alphabetic suffix (maximum 2 characters).

Isotope and element analyses are routinely conducted using the GeoHistory laser ablation ICP-MS and Sensitive High-Resolution Ion Microprobe (SHRIMP) facilities at the John de Laeter Centre (JdLC), Curtin University, with the financial support of the Australian Research Council and AuScope National Collaborative Research Infrastructure Strategy (NCRIS), and the Microscopy Australia laser ablation ICP-MS facility at the Centre for Microscopy, Characterisation & Analysis (CMCA), The University of Western Australia (UWA). The TESCAN Integrated Mineral Analyser (TIMA) instrument was funded by a grant from the Australian Research Council (LE140100150) and is operated by the JdLC with the support of the Geological Survey of Western Australia, UWA and Murdoch University. Mineral analyses are routinely obtained using the electron probe microanalyser (EPMA) facilities at the CMCA at UWA, at Adelaide Microscopy at the University of Adelaide and the Central Science Laboratory at the University of Tasmania.

Digital data related to WA Geology Online, including geochronology and digital geology, are available online at the Department's Data and Software Centre and may be viewed in map context at GeoVIEW.WA. For further information about this Geochronology Record, please contact geochronology@dmirs.wa.gov.au and quote the geochronology record and/or the sample number.

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